Mapping the land-use footprint of Brazilian soy embodied in international consumption: a spatially explicit input–output approach

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Abstract

TODO — Methods-only working document. The full abstract lives in the main manuscript. Placeholder so the file compiles standalone.

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1 Methods

1.1 Overview and notation

We couple a subnational (municipality-level) reconstruction of Brazil's soy supply chain with the global FABIO physical multi-regional input–output (MRIO) framework [1], and attribute the resulting land-use footprint back to municipalities of origin and forward to countries of final consumption, annually for 2010–2020. The construction proceeds in six stages: (i) municipal soy accounting, assembling production, trade and domestic-use quantities for every Brazilian municipality (Section 1.2); (ii) livestock-system disaggregation and feed demand (Section 1.3); (iii) reconciliation of the municipal totals with national commodity balances (Section 1.4); (iv) allocation of physical flows between municipalities by cost-minimizing optimal transport (Section 1.5); (v) linkage of origin municipalities to importing countries, with re-export disentangling (Section 1.6); and (vi) nesting into FABIO and land-use footprint accounting (Sections 1.7–1.8). The resulting flows are validated against the independent Trase supply-chain benchmark and a pure-downscaling baseline (Section 1.9).

The subnational tracing framework generalizes the single-year (2013) open model of Trsek [2] to an annual series; a script-level mapping of each subsection to the released pipeline code is provided with the repository. Figure 1 summarizes the full pipeline.

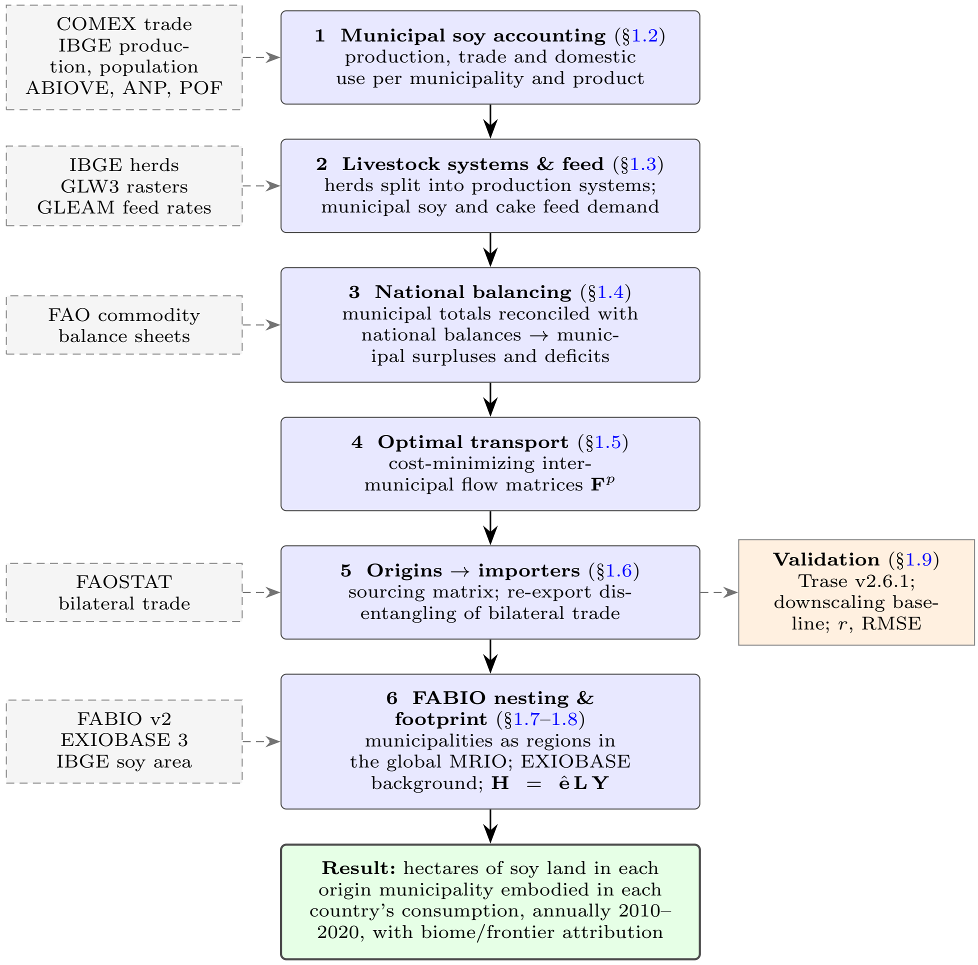

Fig. 1 Workflow of the study. Solid boxes are the six model stages (Sections 1.2–1.8); dashed boxes are the main data inputs at each stage; the orange box is the independent validation of the reconstructed flows (Section 1.9). Stages 1–5 reconstruct Brazil's subnational soy supply chain; stage 6 embeds it in the global FABIO/EXIOBASE input–output framework and computes the land-use footprint.

Notation.

Throughout, m and n index Brazil's ∼5,570 municipalities, $p \in \{\text{bean}, \text{oil}, \text{cake}\}$ the three soy products, and i, j countries. National annual totals from the FAO commodity balance sheet are written $\text{CBS}[p, \text{use}]$ — e.g. $\text{CBS}[\text{oil}, \text{food}]$ for the national food use of soy oil. Matrices and vectors are set in bold, diag(·) denotes a diagonal matrix and $\mathbf{I}$ the identity; a hat ($\hat{\mathbf{e}}$) diagonalizes a vector.

The mass-conserving allocation operator.

A single operation recurs throughout the subnational reconstruction: a national total T is distributed across municipalities in proportion to an observable weight w_m,

$$q_m = \frac{w_m}{\sum_n w_n} T, \qquad \text{so that} \qquad \sum_m q_m = T. \tag{1}$$

The stages below differ mainly in the choice of weight w_m; each allocation is verified in code against its national total. Table 1 collects the weights used for the domestic-use items.

1.2 Municipal soy accounting

1.2.1 Municipal supply, trade and infrastructure

Municipality-level exports and imports of the three soy products are read from COMEX customs records, aggregated from monthly to annual tonnages, and matched to IBGE municipality identifiers. Exports declared without a municipality of origin are redistributed across the declared exporters of the same product and destination in proportion to their declared volumes, which conserves national totals by construction; undeclared imports are negligible and not redistributed. Installed crushing and refining capacity (ABIOVE) is divided equally among each state's active plants and summed per municipality; grain-storage capacity is aggregated from georeferenced warehouse records; and biodiesel capacity (ANP) is attributed to soy by each macro-region's soy feedstock share. All geometries are projected to a planar national reference system (SIRGAS 2000 / Brazil Polyconic, EPSG:5880), and each municipality is represented by its seat; straight-line distances between seats later serve as the transport costs of Section 1.5.

1.2.2 Domestic-use estimation

Every domestic-use item of the national commodity balance is distributed across municipalities with the allocation operator (1), using the weights in Table 1. The national soybean crush is allocated in proportion to crushing capacity; municipal oil and cake output then follow from the crush with conversion yields derived each year from the national balance itself (the ratio of national oil and cake production to beans crushed, ≈20% oil and ≈75% cake, with a ≈5% processing loss, consistent with literature values). Food use of beans and oil is allocated by household soy-oil consumption, constructed as the state-level per-capita acquisition rate (POF) times municipal population; other (biodiesel) use by soy-attributable biodiesel capacity; seed use by soybean production; and stock changes by storage capacity. A net national stock *addition* is carried on the use side; a net national *withdrawal* is instead treated as a component of *supply* (Eq. 2), distributed across municipalities by the same storage weight but added to municipal supply rather than subtracted from use. This mirrors the commodity-balance identity, in which a drawdown is a genuine source of current-year supply, and keeps every municipal use non-negative—as required by the re-export inversion of Section 1.6.2, since a storage-based split of a drawdown on the *use* side would instead push the net use of high-storage, low-use municipalities negative.

Table 1 Domestic-use items as instances of the allocation operator (1): each item is a national commodity-balance total distributed across municipalities by the stated weight w_m.

Use item	Allocation weight w_m
Processing (crush)	installed crushing capacity
Food (beans, oil)	per-capita soy-oil acquisition $\times$ population
Other (biodiesel)	soy-attributable biodiesel capacity
Seed	soybean production
Stock change	storage capacity (addition $\rightarrow$ use; withdrawal $\rightarrow$ supply)

1.3 Livestock systems and feed demand

IBGE municipal herd counts are split into production systems — extensive and intensive chicken; extensive, intensive and industrial pig; grassland- and mixed-based cattle and buffalo, each divided into dairy and meat — because soy feed intake differs strongly across systems. System shares are obtained by area-weighted zonal statistics of the 2010 FAO gridded-livestock rasters (GLW3) and the GLEAM ruminant production-system map over each municipality; where a municipality reports a herd but had no gridded animals in 2010, the state-mean share is used. Dairy herds are identified by milked-cow counts, and feedlot cattle — reported only in the agricultural census — are extrapolated from 2006 with the national confinement growth rate while holding the municipal pattern fixed. The full split formulas are given in Appendix A; each species' subsystems are verified to partition its reported herd exactly.

Feed demand follows by multiplying each herd by its GLEAM per-animal intake: annual dry-matter intake times the share sourced from soybeans and from soy cake, converted to as-fed tonnes at 88% dry-matter content. A single national factor then rescales the resulting bottom-up totals to the FAO national feed figures — an application of (1) with the bottom-up municipality×system feed matrix as weight — preserving the spatial and cross-system distribution while matching the national constraint. Municipal feed demand for beans and cake is the sum over systems. Figure 2 sketches the construction.

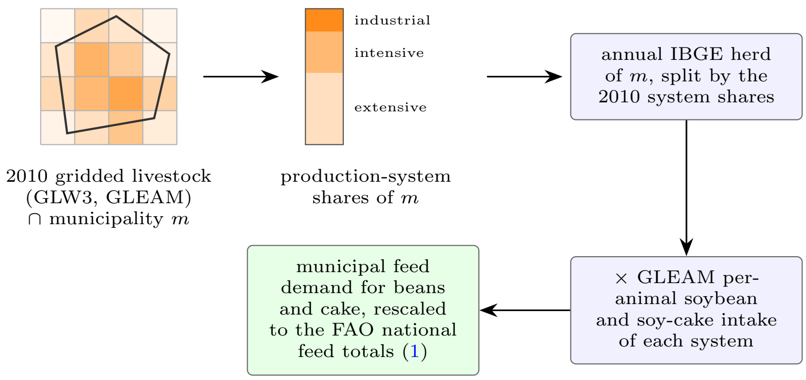

Fig. 2 From herds to municipal soy feed demand. Municipal herd counts (annual) are split into production systems using the shares implied by the 2010 gridded-livestock rasters within each municipality; each system's herd is multiplied by its per-animal soy and soy-cake intake, and the bottom-up totals are rescaled to match the national feed use reported by FAO.

1.4 National balancing and trade reconciliation

For each product, the national commodity balance imposes the supply–use identity

$$\begin{aligned}&\underbrace{\text{production} + \text{imports} + \text{stock withdrawal}}_{\text{supply}}\\ &\qquad = \underbrace{\text{exports} + \text{food} + \text{feed} + \text{seed} + \text{processing} + \text{other}}_{\text{use}}.\end{aligned} \tag{2}$$

Municipal exports and imports are first harmonized with the FABIO/FAOSTAT bilateral trade classification (partners absent from the bilateral trade list are collapsed into a rest-of-world aggregate); this step aligns classifications and applies no rescaling. Any remaining supply–use residual at the national level is absorbed into the stock term, and each balance item is then proportionally rescaled — again (1), with the item's municipal distribution as weight — so that every municipal aggregate matches its national total exactly. Consistent with the supply–use identity (2), a net national stock withdrawal is carried on the *supply* side: the drawdown is distributed across municipalities by storage capacity and added to municipal supply, so that every municipal use remains non-negative (addition years, carried on the use side, are unchanged). Because the model is a single annual snapshot, tonnes supplied from a drawdown—physically grown in earlier years—inherit the current year's soy-area intensity (Section 1.8); this approximation affects only the small share of throughput met from net stock changes. Finally, each municipality's product balance defines its surplus or deficit,

$$\text{surplus}^p_m = \max\big(\text{supply}^p_m - \text{use}^p_m,\, 0\big), \qquad \text{deficit}^p_m = \max\big(\text{use}^p_m - \text{supply}^p_m,\, 0\big), \tag{3}$$

which seed the inter-municipal flows of Section 1.5.

1.5 Subnational transport allocation

Municipal surpluses and deficits are connected by *cost-minimizing optimal transport* — not a gravity or distance-decay model. For each product p, the inter-municipal flow matrix $\mathbf{F}^p$ solves the balanced transportation problem

$$\mathbf{F}^p = \arg\min_{\mathbf{F}\geq 0} \sum_{m,n} F_{mn}\, d_{mn} \quad \text{s.t.} \quad \sum_{n} F_{mn} = \text{surplus}^p_m, \quad \sum_{m} F_{mn} = \text{deficit}^p_n, \qquad (4)$$

where d_{mn} is the straight-line distance between the seats of municipalities m and n (the two margins are balanced beforehand by proportionally scaling the larger one). The problem is solved exactly by the network-simplex algorithm (R package `transport` [3]), separately for beans, oil and cake. The solution is a deterministic, mass-conserving flow matrix whose margins reproduce the municipal surpluses and deficits by construction; its adequacy is assessed against the independent Trase benchmark in Section 1.9. Figure 3 illustrates how municipal balances turn into inter-municipal flows.

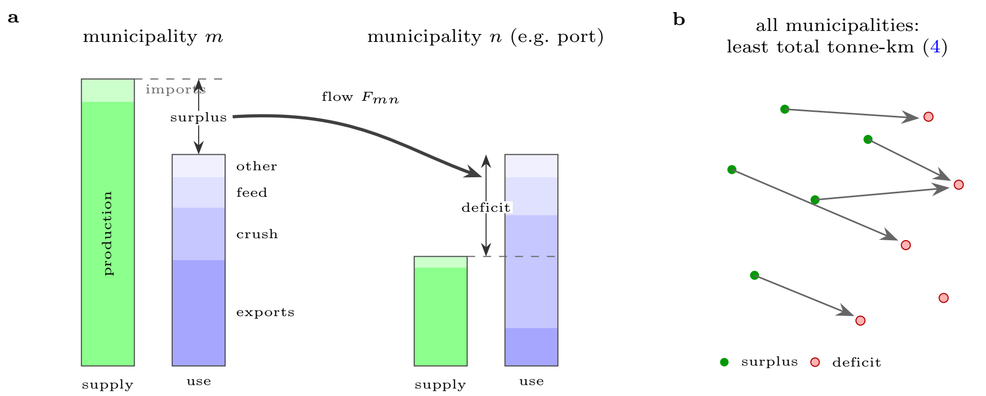

Fig. 3 From municipal balances to inter-municipal flows. **a**, After balancing, each municipality's supply (production, imports) and use (exports, crush, feed, food, seed) differ by a surplus or a deficit (Eq. 3). Deficit municipalities are typically ports and crushing hubs, whose exports and processing exceed local production. **b**, Optimal transport connects all surpluses to all deficits with the least total tonne-kilometres, yielding the flow matrix $\mathbf{F}^p$ for each product; on straight-line costs an optimal plan contains no crossing flows.

1.6 From origins to consumers

1.6.1 Linking origin municipalities to importing countries

The flow matrix is converted into *sourcing shares*: the fraction of municipality n's throughput that originated in municipality m,

$$\phi^p_{mn} = \frac{F^p_{mn}}{\sum_{m'} F^p_{m'n}}. \qquad (5)$$

Combining the sourcing shares with the observed exports E^p_{nj} of municipality n to country j (with a synthetic domestic column carrying local consumption) attributes every exported tonne to its municipality of origin,

$$X^p_{mj} = \delta^p_m \sum_n \phi^p_{mn} E^p_{nj}, \tag{6}$$

where δ^p_m — the share of municipality m's supply that is its own production — retains only domestically produced quantities. Oil and cake are traced one step further back to the bean-producing municipalities by applying the bean sourcing shares to their crush inputs.

1.6.2 Re-export disentangling

To attribute internationally traded soy to its true origin, bilateral trade is disentangled with a Leontief-type inversion. For each commodity, let $\mathbf{T}$ be the bilateral trade matrix — with Brazil replaced by its municipalities for the soy products — and $\mathbf{u}$ the vector of total use by region. Normalizing trade by destination use and summing the infinite re-export chain,

$$\mathbf{R} = \mathbf{T}\,\hat{\mathbf{u}}^{-1}, \qquad \mathbf{L}_{\text{trade}} = (\mathbf{I} - \mathbf{R})^{-1} = \sum_{k=0}^{\infty} \mathbf{R}^k, \tag{7}$$

and retaining only domestically produced quantities (via each region's domestic-supply share) scaled by each destination's domestic use, yields the origin-to-consumption matrix whose entry gives the quantity produced in region (or municipality) i and ultimately consumed in region j. By construction its row and column sums recover each region's domestic supply and domestic use.[1] Figure 4 illustrates the correction for a stylized example.

1.7 Nesting into FABIO

The reconstructed municipal soy flows are embedded in the FABIO v2 physical MRIO, in which a *process* is a region×commodity unit. Brazil's national soy commodities and processes are removed and replaced by their municipal counterparts throughout, so that municipalities enter the global model as additional regions. Figure 5 illustrates the nesting and the role of the Leontief inverses used below.

1.7.1 Supply and use tables

The supply side follows the standard FABIO construction [1]: joint production is split by process production shares, and a parallel monetary table is obtained from bilateral-trade unit prices (winsorized to each item's 10th–90th percentile and gap-filled by a

[1] For the soy products the region dimension comprises non-Brazilian trade countries plus Brazilian municipalities, with national Brazil fully replaced by its municipalities. The implementation re-asserts this replacement after every trade-data join and verifies the dimensional alignment of the inversion explicitly, so that no national-Brazil record can re-enter.

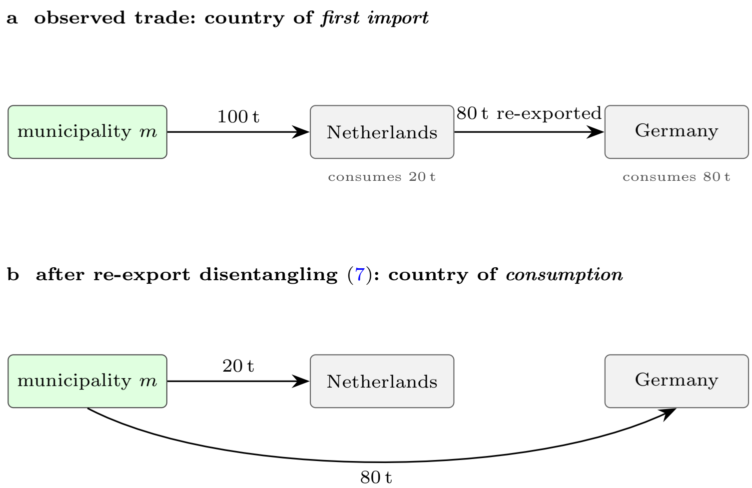

Fig. 4 Re-export disentangling. **a**, Customs records attribute all 100 t exported by municipality m to the Netherlands, the country of first import, although 80 t are re-exported and consumed in Germany. **b**, The trade inversion (7) sums re-export chains of any length and reattributes each tonne to its country of final consumption, leaving only the 20 t actually consumed in the Netherlands attributed there.

world-price, then item-median cascade), with every municipality inheriting Brazil's national price. On the use side, the municipal soybean and soy-cake feed demands of Section 1.3 replace the national feed use, aggregated to the livestock-husbandry processes of each municipality. All other use categories (processing conversion, seed, losses, other uses) are unchanged from the national model.

1.7.2 Multi-regional linking

Each region's supply forms one block of a block-diagonal multi-regional supply table $\mathbf{V}$; the soy blocks are municipal. The re-export-free trade matrix of Section 1.6.2 provides the *trade shares* — for each destination region r and commodity i, the fraction sourced from each origin r',

$$\sigma_{(r',i),\,r} = \frac{\mathrm{trade}_{r'\to r,\,i}}{\sum_{r''} \mathrm{trade}_{r''\to r,\,i}}, \tag{8}$$

which distribute each region's intermediate use and final demand across origins (grazing and stock withdrawals are forced fully domestic). Municipal livestock use and final demand are then re-aggregated to Brazilian national columns, while municipal soy *production* stays disaggregated — the municipal detail is retained exactly where the footprint originates.

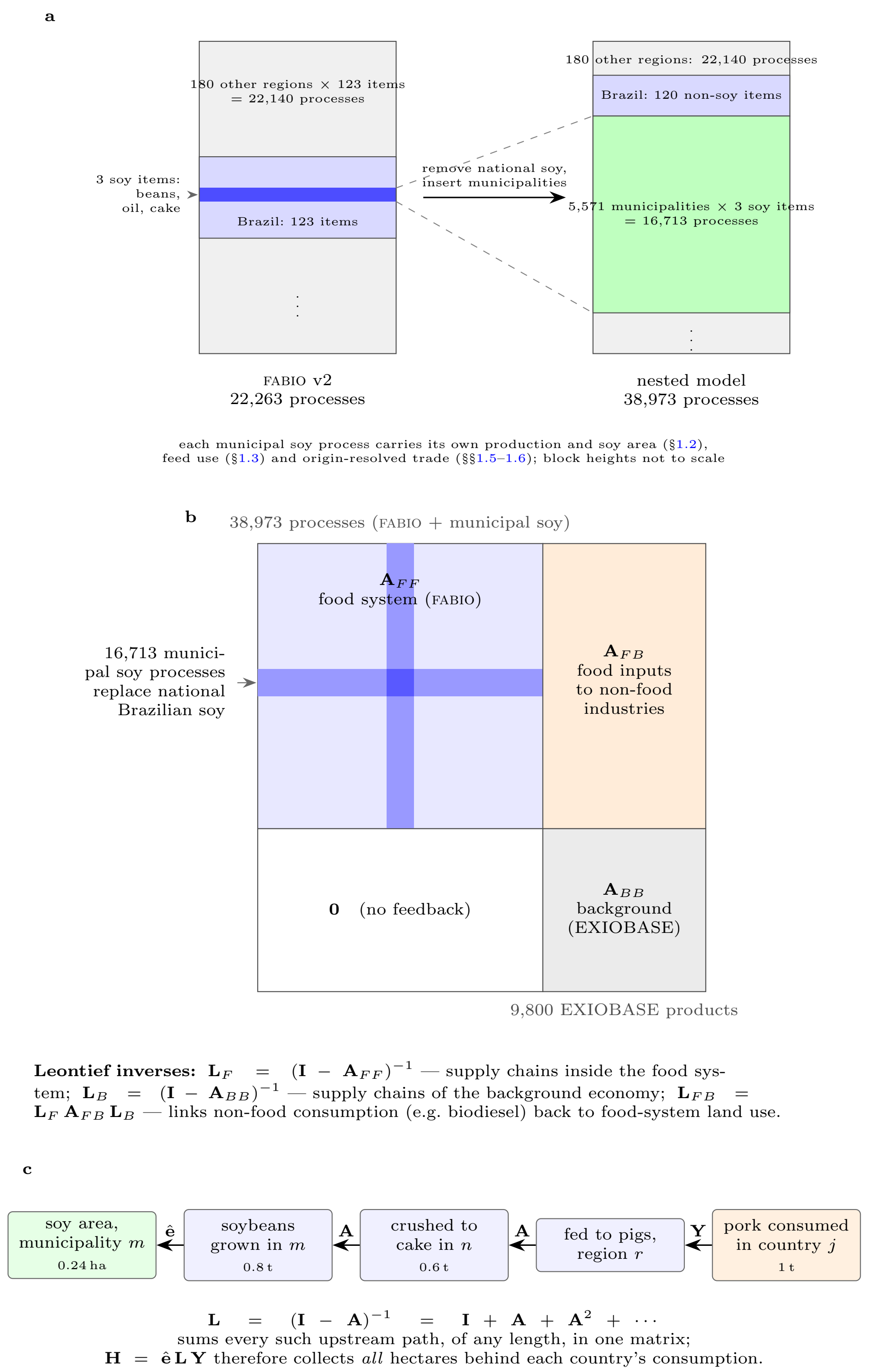

Fig. 5 Nesting municipal soy into the global input–output system. **a**, Insertion of municipal Brazil into FABIO: the three national Brazilian soy commodities are removed from the process list and replaced by one soy process set per municipality, populated with the municipal accounts of Sections 1.2–1.6; all other regions, and Brazil's remaining commodities, are untouched. **b**, Structure of the hybrid technical-coefficient matrix. Brazilian municipalities enter the FABIO block $\mathbf{A}_{FF}$ as additional regions (dark band), replacing national Brazilian soy; the EXIOBASE background economy $\mathbf{A}_{BB}$ receives food-system inputs through the linking block $\mathbf{A}_{FB}$ but does not feed back, so the system is block-triangular. **c**, What the Leontief inverse does: tracing one example consumption–production chain backwards from final demand ($\mathbf{Y}$) through intermediate inputs ($\mathbf{A}$) to the land intensity of the origin municipality ($\hat{\mathbf{e}}$); quantities are illustrative (0.6 t cake per tonne of pork, 0.8 t beans per 0.6 t cake, 3.3 t ha^{-1} yield).

1.7.3 Input–output system and Leontief inverse

The multi-regional supply and use tables are condensed into a symmetric commodity-by-commodity MRIO under the industry-technology assumption [4], and the production chain is summed by the Leontief inverse (Fig. 5c):

$$\mathbf{Z} = \mathbf{U}\,\mathrm{diag}(\mathbf{V1})^{-1}\mathbf{V}, \qquad \mathbf{A} = \mathbf{Z}\,\mathrm{diag}(\mathbf{x})^{-1}, \qquad \mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}, \tag{9}$$

where $\mathbf{U}$ is intermediate use, $\mathbf{Z}$ the resulting transaction matrix and $\mathbf{x}$ total output (equal to total supply; any residual is routed to a dedicated balancing final-demand column). The inverse is computed by sparse LU factorization, in parallel on a physical (mass) and a monetary (value) basis, which differ only in the split of joint production. Spurious negative coefficients arising from the balancing residual are set to zero; a numerical safeguard against singular $\mathbf{I} - \mathbf{A}$ is never triggered in the 2010–2020 study period.

1.7.4 Hybridization with EXIOBASE

Non-food uses of soy (e.g. biodiesel, oleochemicals) leave the food system and enter the wider economy. This background is supplied by EXIOBASE 3 [5] (product-by-product, 49 regions × 200 products): binary concordances map FABIO commodities and countries to EXIOBASE products and regions, and each destination's "other use" is split across EXIOBASE sectors in proportion to that region's product technology, yielding a linking coefficient block $\mathbf{A}_{FB}$. Since there is no physical feedback from the background economy into FABIO, the hybrid system is block-triangular and its cross-system Leontief block is simply

$$\mathbf{L}_{FB} = \mathbf{L}_F\,\mathbf{A}_{FB}\,\mathbf{L}_B, \tag{10}$$

with $\mathbf{L}_F$ the FABIO and $\mathbf{L}_B$ the EXIOBASE Leontief inverse; $\mathbf{L}_{FB}$ carries final demand for non-food products back to FABIO land use.

1.8 Land-use footprint accounting

1.8.1 Footprint calculation

The direct land use of each process is its harvested soy area (from IBGE, for municipal soy processes) or the FABIO cropland extension (for national processes); dividing by total output gives the land intensity $e_i = \mathrm{land}_i/x_i$ of every process i. The footprint is the standard intensity–Leontief–demand product (Fig. 6),

$$\mathbf{H} = \hat{\mathbf{e}}\,\mathbf{L}\,\mathbf{Y}, \tag{11}$$

where $\mathbf{Y}$ is final demand aggregated to consumer countries and H_{ij} is the hectares of soy land in origin municipality i embodied in country j's consumption.[2] The food

[2]Stock additions and the balancing category are removed from final demand before aggregation: stock drawn down in year t embodies land use of earlier production years, and retaining it as negative year-t demand would attribute negative footprints to producer regions in drawdown years. EXIOBASE inventory changes are treated analogously.

system ($\mathbf{L}_F$) and the non-food background (via $\mathbf{L}_{FB}$) contribute additively to each consumer's total. Footprints by consumed product are obtained by restricting $\mathbf{Y}$ to selected commodity groups (e.g. dairy, meat) before applying (11).

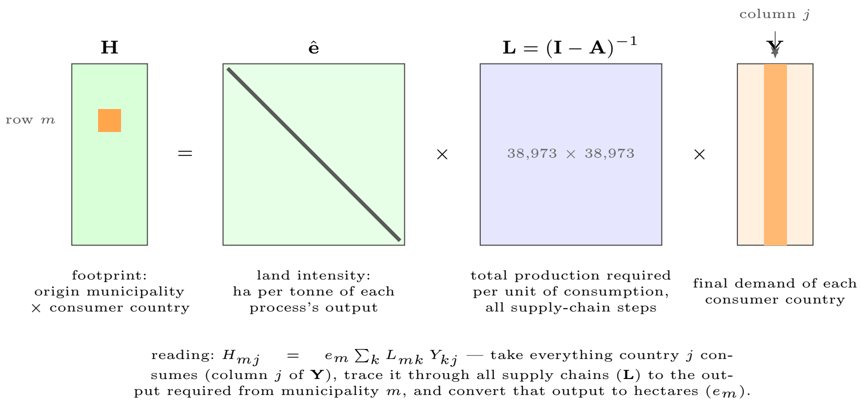

Fig. 6 Linking consumption to land use. The footprint matrix $\mathbf{H}$ is the product of three objects: the land intensity of every process ($\hat{\mathbf{e}}$), the Leontief inverse summing all direct and indirect production requirements ($\mathbf{L}$), and final demand by consumer country ($\mathbf{Y}$). The highlighted entry H_{mj} is the soy area in municipality m embodied in country j's consumption.

1.8.2 Origin attribution and grid refinement

Each municipality is assigned to a biome by intersecting its interior point with the IBGE 1:250,000 biome layer. The agricultural *frontier* is defined as the Amazon biome plus MATOPIBA (operationalized as the Cerrado biome within Maranhão, Tocantins, Piauí and Bahia), and a destination's frontier share is the fraction of its origin footprint located in frontier municipalities. For visualization, per-municipality embodied production can optionally be distributed onto 30 m MapBiomas soy pixels, each pixel of a municipality carrying that municipality's per-pixel share for the given consumer country; this refinement is illustrative and not load-bearing for the results.

1.9 Validation and benchmarking

1.9.1 Benchmark and baseline

The reconstructed origin-to-importer flows X^p_{mj} (Section 1.6.1) are evaluated against the independent Trase Brazil–soy dataset (v2.6.1), which links municipalities of production to countries of first import using per-shipment customs and logistics records [6, 7]. As a second reference we construct a *pure-downscaling* baseline that uses no supply-chain structure at all: national exports to each destination are allocated to municipalities in proportion to their share in national soybean production — the allocation operator (1) applied directly to bilateral exports. The comparison thus brackets the model between an information-free lower bound and a data-rich proprietary benchmark. All three sources are expressed in a common unit by converting oil and cake flows to bean equivalents with the national crush ratio (beans crushed per tonne of oil and cake produced). Trase flows with unknown municipality of origin are excluded, destinations are harmonized to the bilateral-trade classification, and the comparison sample comprises all soy-producing municipalities crossed with the destinations recording positive exports in both benchmark and model.

1.9.2 Agreement metrics

Agreement is quantified on the municipality×destination flow matrix by the Pearson correlation coefficient r (with Fisher 95% confidence intervals), the root-mean-square error and the root-mean-square logarithmic error, computed (i) pooled over all pairs, (ii) per destination country, and (iii) after aggregation to states. Because a municipality-level assignment can be "nearly right" — placing a flow in the neighbouring municipality of the same production cluster — all metrics are additionally computed on neighbourhood-smoothed flows, using three spatial-weight schemes (first-order contiguity, a 100 km radius, and inverse distance, each raw and row-standardized). Robustness of the conclusions across these tolerances distinguishes genuine misallocation from boundary-precision noise.

1.9.3 Volume–correlation decision rule

Per-destination correlations are related to the destination's total import volume to establish when subnational modelling adds information over pure downscaling. Small trade volumes are handled by few shipments and carry no recoverable spatial signal, so per-destination agreement degrades for all methods below a volume floor (on the order of 10^3 t; Results). The resulting decision rule — model subnationally above the floor, downscale below it — is reported with the benchmark tables produced for every study year.

A Livestock system splits

Let s_m^k denote municipality m's share of system k in the 2010 gridded-livestock rasters (area-weighted zonal sums, normalized across the systems of each species); where a municipality reports a herd in the run year but had no gridded animals in 2010, the state-mean share is substituted. Pigs and chickens are split as

$$N_m^{\mathrm{pig},k} = \mathrm{pig}_m\, s_m^{\mathrm{Pg},k}, \quad N_{m,\mathrm{byd}}^{\mathrm{chk}} = \mathrm{chicken}_m\, s_m^{\mathrm{Ch,Ext}}, \quad N_{m,\mathrm{lay}}^{\mathrm{chk}} = \mathrm{chicken\_layer}_m\, s_m^{\mathrm{Ch,Int}}, \tag{12}$$

with broilers $N_{m,\mathrm{bro}}^{\mathrm{chk}} = (\mathrm{chicken}_m - \mathrm{chicken\_layer}_m)\, s_m^{\mathrm{Ch,Int}}$ (or $\mathrm{chicken}_m\, s_m^{\mathrm{Ch,Int}}$ where layers are unreported). Feedlot cattle are the 2006 census counts scaled by the national confinement growth factor for the run year (e.g. $4.38/3.46 \approx 1.27$ for 2013, from ABIEC/Athenagro), holding the municipal pattern fixed. The remaining cattle are split into grassland/mixed systems by the raster shares $s_m^{\mathrm{Catt},u}$, $u \in \{\mathrm{gra}, \mathrm{mix}\}$, with dairy identified by milked cows and meat as the residual:

$$N_{m,u,\mathrm{dair}}^{\mathrm{cattle}} = \mathrm{cattle\_milked}_m\, s_m^{\mathrm{Catt},u}, \tag{13}$$

$$N^{\text{cattle}}_{m,u,\text{meat}} = \left(\text{cattle}_m - \text{cattle\_milked}_m - N^{\text{cattle,flot}}_m\right) s^{\text{Catt},u}_m. \tag{14}$$

Buffalo use the municipal milked-cow ratio $\rho_m = \text{cattle\_milked}_m/\text{cattle}_m$ as the dairy proxy,

$$N^{\text{buf}}_{m,u,\text{dair}} = \text{buffalo}_m\, \rho_m\, s^{\text{Buff},u}_m, \qquad N^{\text{buf}}_{m,u,\text{meat}} = \text{buffalo}_m\, (1-\rho_m)\, s^{\text{Buff},u}_m. \tag{15}$$

Each species' subsystems partition its reported herd exactly, e.g. $\text{cattle}_m = N^{\text{cattle,flot}}_m + \sum_u \sum_{v\in\{\text{dair,meat}\}} N^{\text{cattle}}_{m,u,v}$.

References

[1] Bruckner, M. *et al.* FABIO—the construction of the food and agriculture biomass input–output model. *Environmental Science & Technology* **53**, 11302–11312 (2019).

[2] Trsek, S. *Refining footprint analyses beyond the national level: An open tracing framework to capture the heterogeneity of subnational commodity flows.* Master's thesis, University of Natural Resources and Life Sciences (BOKU), Vienna (2022).

[3] Schuhmacher, D. *et al.* transport: Computation of optimal transport plans and Wasserstein distances (2023). URL https://cran.r-project.org/package=transport. R package.

[4] Miller, R. E. & Blair, P. D. *Input-Output Analysis: Foundations and Extensions* 2 edn (Cambridge University Press, Cambridge, 2009).

[5] Stadler, K. *et al.* EXIOBASE 3: Developing a time series of detailed environmentally extended multi-regional input-output tables. *Journal of Industrial Ecology* **22**, 502–515 (2018).

[6] Godar, J., Persson, U. M., Tizado, E. J. & Meyfroidt, P. Towards more accurate and policy relevant footprint analyses: Tracing fine-scale socio-environmental impacts of production to consumption. *Ecological Economics* **112**, 25–35 (2015).

[7] Escobar, N. *et al.* Spatially-explicit footprints of agricultural commodities: Mapping carbon emissions embodied in Brazil's soy exports. *Global Environmental Change* **62**, 102067 (2020).